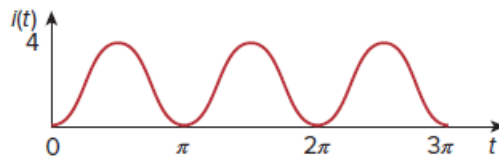


Problem

Find the rms value of the offset sine wave shown in Fig. 11.53.

Figure 11.53:



Step-by-step solution

Step 1 of 3

Refer to Figure 11.53 in the textbook for the offset sine wave.

The waveform is a addition of a dc voltage of 2 V and a sinusoidal voltage of $-2\cos 2t$ V. The ac and dc components are shown separately in Figure 1.

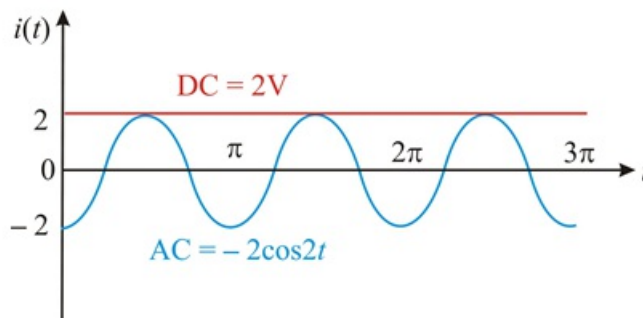


Figure 1

Step 2 of 3

The mathematical expression for current $i(t)$ is,

$$i(t) = 2 - 2\cos 2t \quad 0 < t < \pi$$

Calculate the root mean square (rms) value of the current $i(t)$.

$$\begin{aligned} I_{\text{rms}}^2 &= \frac{1}{T} \int_0^T i^2(t) dt \\ &= \frac{1}{\pi} \int_0^\pi (2 - 2\cos 2t)^2 dt \\ &= \frac{1}{\pi} \int_0^\pi (4 + 4\cos^2 2t - 8\cos 2t) dt \\ &= \frac{1}{\pi} \int_0^\pi \left(4 + 4 \left(\frac{1 + \cos 4t}{2} \right) - 8\cos 2t \right) dt \quad \text{since } \left(\cos^2 x = \frac{1 + \cos 2x}{2} \right) \\ &= \frac{1}{\pi} \int_0^\pi (4 + 2 + 2\cos 4t - 8\cos 2t) dt \\ &= \frac{1}{\pi} \int_0^\pi (6 + 2\cos 4t - 8\cos 2t) dt \end{aligned}$$

Step 3 of 3

Simplify further.

$$\begin{aligned} I_{\text{rms}}^2 &= \frac{1}{\pi} \left[6t + 2 \frac{\sin 4t}{4} - 8 \frac{\sin 2t}{2} \right]_0^\pi \\ &= \frac{1}{\pi} \left[6(\pi - 0) + \frac{(\sin 4\pi - \sin 0)}{2} - 4(\sin 2\pi - \sin 0) \right] \\ &= \frac{1}{\pi} (6\pi) \quad \text{since } (\sin 4\pi = \sin 2\pi = 0) \\ &= 6 \end{aligned}$$

Now, calculate the value of I_{rms} .

$$\begin{aligned} I_{\text{rms}} &= \sqrt{6} \\ &= 2.45 \text{ A} \end{aligned}$$

Therefore, the rms value, I_{rms} of the offset sine wave is 2.45 A.